The Cancer Vaccine That Could Save Moderna

Podcast: Wall street journal

AI Executive Summary

1. Core Thesis Despite hemorrhaging cash and facing a stock price collapse of over 90% since its pandemic peak, Moderna has pivoted from its infectious disease focus to oncology, announcing promising clinical data for an mRNA melanoma cancer vaccine that stops tumor growth and metastasis. This breakthrough represents a potential company turnaround by validating their core technology in a high-growth market, with the CEO declaring that Moderna is now officially an oncology company following earlier phase two success.

2. Key Arguments & Data

3. **Financial Performance:** Moderna stock has dropped more than 90% from its peak; the company has been "bleeding cash" for over two and a half years with short interest hitting 20%.
4. **Revenue History:** The COVID-19 vaccine generated $36 billion in sales over two years, which funded expansion of other programs including oncology.
5. **Clinical Trials (Oncology):**
    - Phase 2 Study (2022): Interim data met two goals—preventing growth and preventing metastasis (spreading to other organs) in melanoma patients receiving the combination therapy with Merck’s Ketruda.
    - Phase 3 Study: Over 1,000 high-risk melanoma patients were enrolled; the study concluded sooner than expected.
    - Results: The combination appeared to stop cancer from returning or spreading in high-risk patients. Specific efficacy percentages are pending a presentation at a medical conference later this year.
6. **Partnership:** Moderna partnered with Merck, which provides Ketruda (immunotherapy drug) for the trial and distribution support.
7. **R&D Cost & Timeline:**
    - CEO Stefan Boncel previously cited cancer research as too expensive ($500k+ per project initially deemed too risky).
    - Moderna was awarded over $1 billion in funding from the Trump administration's Operation Warp Speed for the COVID vaccine,

enabling rapid development.
    - Future pricing is expected to be less than $500,000 per patient, significantly lower than other oncology therapies.

8. **Strategic Shift:** Originally focused on flu and genetic diseases (2010s), Moderna pivoted to cancer after investor Stephen Hogue funded an initial 2013 grant ($500k check) despite CEO resistance due to high costs.

9. Actionable Takeaways

10. **Monitor Specific Data:** Investors should await the specific efficacy percentages reported at the upcoming medical conference later this year, as current data is qualitative rather than quantitative.
11. **Watch FDA Approval:** The company targets FDA approval next year; regulatory scrutiny on mRNA technology for cancer remains a critical risk factor alongside the previous skepticism of the Trump administration regarding the technology.
12. **Investment Caution vs. Opportunity:** While Moderna represents high-risk, high-reward potential due to its pivot into oncology (a massive market), investors must acknowledge the company has historically struggled with cash reserves before the pandemic windfall.
13. **Understand the Model:** Recognize that Moderna is transitioning from a platform biotech focused on mRNA delivery for infectious diseases to a therapeutic oncology player requiring tailored treatments (bespoke vaccines).

Full Transcript

The biotech company Moderna is best known for its COVID vaccine. The vaccine put the small startup on the map and made it billions of dollars. But in recent years, its business has been in deep trouble. COVID starts to sort of wind down in terms of the urgency that we're treating it with. Sales start to drop in the US because not as many people are getting vaccinated. That's our colleague Xavier Martinez, who covers biotech. And before you know it,

Their stock has really bottomed out. It's gone down more than 90% from its peak. The company is bleeding cash. You look at the stock chart and it just looks like the tallest mountain in the world. But earlier this month, Moderna made a surprising announcement. I remember where I was. I was on a call with a different company, with a different pharmaceutical company. earlier this month and I was in the office in a little meeting room and my editor comes over and starts waving her hands up and down and clearly something is happening. Hang up the phone now. There's breaking news. Exactly, exactly. And she starts messaging me and she starts saying like, okay, okay, I really need you to get on a call in five minutes. On the call, Xavier learned that a cancer vaccine that Moderna researchers had been working on showed real promise in clinical studies.

That study met two goals in both preventing the growth of any cancer, but also preventing it spreading from other organs in the body or metastasizing. The vaccine is for people who've already had melanoma, a serious form of skin cancer. It's designed to prevent the cancer from returning or spreading to other parts of the body. How big of a deal potentially is this melanoma vaccine that Moderna has created? It could be one of the biggest advancements in cancer that we've seen in the last 25 years. If Moderna continues down this path that they're going down, this could really just kind of totally change how we treat cancer. Not only that, but it's also breathed new life into a company that some thought was on death's door. This is really bringing Moderna back from the brink financially. They've been hemorrhaging cash for the last two and a half years.

And so I think what it means is, and this is certainly a new chapter for them. They're cancers in remission. Exactly, exactly. I think that's a good way to put it, right? Welcome to The Journal, our show about money, business, and power. I'm Ryan Knudsen. It's Friday, August 28th. Coming up on the show, the cancer vaccine could save lives and the company behind it. A decade before the pandemic, Moderna was just another biotech startup outside of Boston. Moderna was founded in 2010, and it was what we call in the biopharma industry a platform company. And so what that means is that it was designed with the goal that all of its drugs that it would make over the course of its whole lifetime would be all focused on mRNA. mRNA, remind us what that is? mRNA is short for messenger RNA, sort of a sister to regular typical DNA. It's a natural molecule in your body that acts like a sort of a temporary instruction manual or a blueprint. And its job is just to shuttle directions from your DNA to different parts of your body. mRNA carries instructions that teach your body how to make proteins like antibodies to fight diseases. The idea Moderna was founded on was simple. What if scientists write that instruction manual? What if they could engineer mRNA to tell the body how to make antibodies?

And so the idea with this is that you could manipulate the RNA in such a way that it can create antibodies that can fight diseases. Yeah, I mean the goal here is that you can actually encode like different sets of instructions to tell your immune system to say, hey, when you see a mutation like this on a cancer tumor, you should go fight it. But actually writing those instructions is easier said than done. In the early days, did Moderna focus its technology on cancer? Not at all. Cancer... We still don't fully understand, right? I mean, we know that it definitionally is the uncontrolled growth of cells, but we're still learning every day what that actually means from a biological level, right? And so I think it was just a really unproven bet in terms of using mRNA to target any one component of the immune system, right? So early on, the company decided to focus on illnesses that seemed easier to target.

Here's Moderna's CEO, Stefan Boncel, talking about this decision almost a decade ago. Because if we pick the wrong biology, it might not be the technology of the issue, it might be the biology. But everybody, including us, is going to believe it's the technology. Instead of taking on cancer,

Moderna decided to focus on treating things like the flu and genetic diseases. We are very worried in the early days, kind of paranoid worried, of going the wrong way. and taking so long to realize we are in the wrong way on the science that we might kill the company. was very much the realist in the situation here, right? And I think that goes for a lot of the executives. They were very much focused on proving out this technology, showing that it worked, you know, for other sort of types of diseases. And then of course, maybe going into things like oncology later down the line, you know, we were able to sit down with the CEO and some of the executives last week. And they really made a point to say that, no, we were actually kind of afraid of going into oncology too early because it was going to be so cost intensive, it costs millions of dollars to enroll patients in clinical trials and things like that. And more importantly, the science just wasn't there at the time, right? There'd been all these attempts to try to use these therapeutic cancer vaccines to try to reshape the immune system and just none of them really worked. And so, you know, when you're a really small biotech company, you don't have a lot of money, you don't have a lot of runway, you kind of need something that's a little easier to target.

Still, some Moderna executives thought a cancer vaccine was possible and didn't want to give up on that dream, including the company's president, Stephen Hogue. Stephen Hogue, who is a scientist, he's a physician, he was a McKinsey consultant. He's sort of, maybe you could think of him as a little more of the dreamer and the scientist of the executive group. I have come to realize that optimism is my greatest privilege. That's Hogue talking about his impressions of Moderna when he joined the company in 2013. Moderna just kept striking me as odd. It was trying to pursue a phenomenally big idea like sort of farcically big that we're gonna figure out a way to turn medicine into information or information into medicine. I was not totally daunted as an optimist when people told me that was a crazy thing to do. More than a decade ago Developments with some other drugs made by other companies made Hoke think Moderna should give mRNA cancer treatment a serious look. In the early and mid 2010s, you see this big wave of immune focused cancer therapies coming out. You can sort of see and trace out the logic about, well, if we can maybe target the immune system and use mRNA to tell it to actually do different things, to target different things, well, maybe that might actually be useful in oncology.

Hogue showed Boncel scientific studies that mapped out new mRNA concepts in cancer biology. But Boncel told Hogue that Moderna didn't have the money to spend on an oncology program. In those days, Moderna had fewer than 100 employees, and it didn't even have a single product in the market, meaning it wasn't bringing in any revenue. So without more funding, Boncel and the realists of Moderna won the argument. Developing a cancer vaccine was just too expensive. and the payoff too uncertain for a little startup like Moderna to spend its limited resources on. But then in 2013, one of Moderna's early investors reached out to Boncel. His wife had been diagnosed with stage four lung cancer. He personally talks to Stefan, the CEO to Boncel, and he kind of says, hey, is there any chance at all that you guys could potentially look into this whole cancer thing and see if your mRNA Technology might be useful there. And Bunsell, the CEO, says, sorry, no. It's going to be way too expensive even just to hire a couple of researchers. It's going to cost 500,000, half a million dollars, right? And so he goes home. He goes back to his office in Cambridge. And a couple of days later, a FedEx envelope shows up on his desk. And it's a check from that hedge fund manager for 500,000. With that cash, Moderna hired two cancer researchers.

and its oncology program got underway. The hard part about treating cancer, though, is that each person's cancer is different, meaning each mRNA vaccine would have to be tailor-made to each individual patient. And making bespoke vaccines is expensive. And then Sel says, well, it's going to cost a lot of money. So can you figure out how to maybe keep the cost down on that? And so a couple of Moderna employees at the time locked themselves in a room. scribble on this whiteboard, Buncel still has a photo of the whiteboard. It has all these different sort of colors written on it in marker, green and purple and red and pink. And, you know, they actually sort of sketch out how this might work. Once they sort of start sketching it out, once they start using the technologies they have in-house, they realize, well, shoot, we're going to need a whole lot more money if we want to actually test to see if this technology is going to work, you know, not in a lab, not on the whiteboard, but if it's actually going to work.

in patients who have cancer. And for that, Moderna needed a partner. What they do is they end up thinking like, OK, there's all these great

pharmaceutical companies that, A, they know how to do clinical trials. They know how to work with academic centers. They have relationships with them. They know the process for making these experimental medications for testing. They know the distribution. Which ones are good at cancer? And so they approach a couple of companies, they're in conversation with a couple of companies, and ultimately to settle on Merck. Merck, the pharmaceutical giant behind one of the most successful cancer drugs, Ketruda. Unlike chemotherapy treatments, which effectively poison cancer tumors, Ketruda is an immunotherapy drug, meaning it stimulates a person's natural immune system to fight cancer, similar to what Moderna was trying to do with mRNA.

So in 2016 Boncel and Hogue visited Merck to pitch the company on a partnership. And so they go to Merck's offices, they pitch this idea, and they say, you know, we have this idea. We actually want to make a vaccine for cancer. And the Merck executives have tons of questions. And at the end, you know, these Merck executives are still doubtful. They frankly... don't necessarily think it's going to work. And I should also say Moderna is not confident that it will work there for years and years and years and even through this process saying, well, we don't want to hype this up too much. We just want to see if this will yield anything because if it does, it would be very promising. And so Merck expresses some doubt, but ultimately they say, let's do it.

Moderna and Merck started a phase two study and administered the vaccine to a small number of cancer patients. At long last, Moderna would be able to see if its cancer vaccine actually worked. A lot of people didn't even think they would get this far with the cancer vaccine, let alone with the mRNA technology. But then something got in the way. The coronavirus outbreak has been declared a pandemic. How COVID-19 nearly derailed the cancer shot is next. Six months after Moderna started the phase two study of its cancer vaccine, the COVID-19 pandemic hit. The world is scrambling to stop the spread of COVID-19. More and more cases are popping up in countries far from the epicenter of the outbreak. And Moderna pivoted quickly to develop a vaccine. Moderna is one of the first companies to develop a vaccine. They do it from when they start development of that vaccine to when they actually get it approved by the FDA on this sort of

expedited pathway as part of Operation Warp Speed at the end of the first Trump administration, it's less than a year, right? And we compare that to other drug development timelines. That's extraordinarily quickly. So I think, you know, puts them on the map, both to the public, but also to investors and to the scientific community saying, Oh, wow, this biotech and actually they can play ball, right? They can put a drug on the market, they can use this mRNA technology to actually address a pressing need that we have here in society, and they can do it very quickly. But as Moderna's priorities shifted to COVID, its cancer program suffered. It is actually very bad for the program. They're using the same equipment that they use for the cancer vaccine. So they literally have to pick, OK, we know we have patients in this clinical trial that have cancer, who might die from that cancer. Are we going to treat them? Are we going to make?

their individualized cancer vaccines, or are we going to make six million doses for the global public? And of course, they chose the latter, right? But talking to the executives, that was not an easy decision. It certainly was not a fun decision to have to make. But that decision led to a huge windfall. Moderna won more than a billion dollars from the Trump administration to speed up its development of its COVID shot, which was authorized for emergency use that December. And in the first two years, the vaccine was on the market. It generated 36 billion in sales. Moderna used that money to expand and grow other programs, including oncology. But as the pandemic subsided, Moderna's business got hit with a hangover. Oh, I mean, a hangover doesn't even begin to describe it. As COVID starts to wind down, as they're losing money, they have to cut programs eventually. They have to cut some of their developmental programs that they're kind of bringing into the clinic to prioritize the ones that they're most excited about.

Moderna cut jobs and ended research programs into things like a pediatric RSV vaccine and other cancer therapies. The stock fell, and some investors even made bets that it would keep falling. Earlier this year, the short interest in Moderna hit 20%. But Moderna's research into a cancer vaccine survived all those cuts. So despite all of the doubts about the viability of a cancer vaccine, why did that research continue? Among all the things that they cut, why didn't they cut that? Well, I think a very important inflection point happens in 2022, which is when they get the results from that phase two

study. Phase two, a critical stage of clinical trials where researchers find out if their cancer vaccine works on actual melanoma patients. The hope was that the vaccine would prevent cancer from returning or spreading in patients diagnosed with melanoma. Phase two studies are really, really important. And for a lot of people in the industry, they sort of represent the make or break.

moment, right? Like you're not going to push something beyond a phase two if you don't really think it's going to read out well. It's going to have some really strong results. And so Moderna and Merck get sort of interim looks at the data in 2022. And the results are really promising for them. Around this time, Merck committed to a full partnership with Moderna to make and sell the cancer vaccine together. Then the study moved to phase three, basically a much bigger version of the phase two study. In both studies, every patient took Merck's existing cancer treatment, Kitruda, but only some also took a dose of Moderna's vaccine. This was a very big trial. It was over 1,000 patients, which, if you think about the logistics of it, meant that it was 1,000 people with high-risk melanoma, getting their melanoma removed, sending that.

and a blood sample to Moderna and Merck, having them actually sequence those mutations, formatting cancer vaccine, and then dosing them for about a year. So throughout the year, Moderna had been signaling that in the second half of this year, in the second half of 2026, they were going to get the results from that phase three, that late stage study of the cancer vaccine in these melanoma patients. And it comes a lot sooner than anyone thinks. The company says the vaccine worked. It was helping real melanoma patients fight cancer. The company announced the high level of results last week. Now according to Mark and Moderna, a combination of the vaccine and medication appeared to stop the cancer from returning or spreading in high-risk patients. Moderna still hasn't shared the specific details about how well the treatment worked and in what percentage of patients. It says it'll present those results at a medical conference later this year.

But the fact that the results came in sooner than expected is a promising sign. Can you describe the scene inside the company when this news spread? It was like your birthday and Christmas and the day you got

married all wrapped up into one. This was a massive, massive bet that Moderna made. This was a very strong indication that it was finally paying off. They sort of all gather on a call and in the headquarters, they pack into their cafeteria. They have champagne and sparkling juice. And I mean, there are people even tuning in from Asia where it's the middle of the night. It's like midnight or 1 AM there. The Moderna CEO is reading off a note card because he doesn't trust himself to sort of not choke up or say things eloquently. This is just everyone is recognizing this is an extraordinary moment. People were crying. People were hugging.

Bunsell, the CEO, put it very plainly, I'd say. He said, we are now an oncology company. For a company that made its name doing infectious diseases and COVID and had all this pipeline of different products that are not at all related to oncology, I think that's a very bold statement to make, but I think it also shows the direction that they're looking in terms of their... He believes this is the future. Exactly, exactly. Still, Moderna and Merck have work to do before the vaccine can be widely administered. For one, it still needs approval from the Food and Drug Administration, and the Trump administration has shown skepticism about mRNA technology. When you talk about treating something with as much of an unmet need as cancer, I think that there is a strong indication that if a technology is effective, it's probably going to be looked fondly upon by the FDA. The companies are hoping for approval next year. And while Moderna's new vaccine shows promise, it's still just in trials.

and it's still just in trials for melanoma. Vaccines for other types of cancer might be possible in the future, but are a long way off. The cost of the vaccine for patients is also unclear. There are other types of cancer therapies, one of which can cost up to 500,000 per patient. Boncel says this new treatment will be priced less than that. This really shows that betting on, again, this sort of thing that the community calls a platform, right? It's a very risky bet. But it can really pay off it if it pays off, right? They're going to have their first entry into the massive oncology market. It can really prove out to be well. And I think it just speaks to their tenacity in being laser focused on the mRNA technology. I mean, these are the experts on it. Before we go, we've got something new for you this weekend. It's called

My Monday Morning, and it's hosted by the Wall Street Journal's Lane Floresheim. Lane, what's the quick pitch?

Well, for the next few Sundays, I'll be talking to someone who's really good at what they do. And we'll discuss something pretty simple. What does your Monday morning actually look like? And it's a surprisingly revealing way to get people to open up about how they think, how they work, and what they've figured out about themselves along the way. Who's your first guest? We're kicking off with one of the most famous financial advisors out there, Susie Orman. And I'll just say, I need a lot more sleep than she does. How much sleep does she get? She says she can make it on as little as three hours a night. Three hours? That's nothing. Who else do you have in the series? I'll be talking to an astronaut, actors, the model Twiggy. It's going to be a lot of fun, and I hope people give it a listen or a watch.

All right, you can watch it because it's also going to be a video on Spotify. Yes, and you can also listen with your ears wherever you get your podcasts. The first episode of my Monday morning drops in the journal feed on Sunday. Thanks, Lane. That's all for today, Friday, August 28th. The journal is a co-production of Spotify and The Wall Street Journal. Additional reporting in this episode by Alex Janin. The show is made by Katherine Brewer, Evelyn Fajardo-Alvarez, Pia Gedkari, Max Green, Sophie Codner, Matt Kwong, Colin McNulty, Jessica Mendoza, Laura Morris, Enrique Perez De La Rosa, Sarah Platt, Alan Rodriguez Espanosa, Heather Rogers, Pierce Singy, Jeevika Verma, Katherine Whalen, Tatiana Zamis, and me, Ryan Knudson. Our engineers are Griffin Tanner, Nathan Singapok, and Peter Leonard. Our theme music is by So Wiley. Additional music this week from Katherine Anderson, Peter Leonard, Bobby Lord, Nathan Singapok, Griffin Tanner, So Wiley, and Brutal Sessions. Fact-checking this week by Nicole Posolka. Thanks for listening. See you on Sunday.